<u>Example</u>: An autonomous taxi in a certain city moves riders between the airport (A), downtown (D), and the city's waterfront area (W).

Let X_n = the location of the taxi at its nth stop

It is believed that $\{X_n\}$ is a Markov chain with transition probability matrix given by

$$\mathbb{P} = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$

Find the following probabilities.

① $P(X_1 = A \mid X_0 = D) = 0.35$

② $P(X_{37} = A \mid X_{36} = D) = 0.35$

③ $P(X_{37} = A \mid X_{36} = D, X_{35} = A)$

$\overset{M.P.}{=} P(X_{37} = A \mid X_{36} = D)$

$= 0.35$

$$\begin{array}{c c} & \begin{array}{ccc} A & D & W \end{array} \\ \begin{array}{c} A \\ D \\ W \end{array} & \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$

④ $P(X_1 = W)$

Not enough info here!

We need an "<u>initial distribution</u>".

Suppose that $P(X_0 = A) = 1/2$, and

$$P(X_0 = D) = P(X_0 = W) = 1/4.$$

We will condition on the value of X_0.

④ $P(X_1 = W)$

$$= P(X_1 = W \mid X_0 = A) \cdot P(X_0 = A)$$
$$+ P(X_1 = W \mid X_0 = D) \cdot P(X_0 = D)$$
$$+ P(X_1 = W \mid X_0 = W) \cdot P(X_0 = W)$$

$$= (0.30)(1/2) + (0.25)(1/4)$$
$$+ (0.15)(1/4) = \boxed{0.25}$$

$$\begin{array}{c c} & \begin{array}{ccc} A & D & W \end{array} \\ \begin{array}{c} A \\ D \\ W \end{array} & \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$

⑤ $P(X_2 = D, X_1 = A, X_0 = A)$

$$= P(X_2 = D \mid X_1 = A, \cancel{X_0 = A}) \cdot P(X_1 = A, X_0 = A)$$

M.P.

$$= P(X_2 = D \mid X_1 = A) \cdot P(X_1 = A \mid X_0 = A) \cdot P(X_0 = A)$$

$$= (0.65)(0.05)(1/2)$$

$$= \boxed{0.01625}$$

$$\begin{array}{c c} & \begin{array}{ccc} A & D & W \end{array} \\ \begin{array}{c} A \\ D \\ W \end{array} & \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$

⑥ $P(X_2 = D \mid X_0 = A)$ ← a "<u>two-step transition probability</u>"

Condition on the value of X_1.

$$= P(X_2 = D \mid X_1 = A, \cancel{X_0 = A}^{\text{M.P.}}) \cdot P(X_1 = A \mid X_0 = A)$$

$$+ P(X_2 = D \mid X_1 = D, \cancel{X_0 = A}^{\text{M.P.}}) \cdot P(X_1 = D \mid X_0 = A)$$

$$+ P(X_2 = D \mid X_1 = W, \cancel{X_0 = A}^{\text{M.P.}}) \cdot P(X_1 = W \mid X_0 = A)$$

$$= (0.65)(0.05) + (0.40)(0.65) + (0.55)(0.30) = \boxed{0.4575}$$

$$P(X_2 = D \mid X_0 = A)$$

$$= (0.65)(0.05) + (0.40)(0.65) + (0.55)(0.30)$$

What do you see?

This is the $A \to D$ entry of $\mathbb{P}^2$.

$$\mathbb{P} = \begin{array}{c} \\ A \\ D \\ W \end{array} \begin{array}{c} \begin{array}{ccc} A & D & W \end{array} \\ \begin{bmatrix} 0.05 & 0.65 & 0.30 \\ 0.35 & 0.40 & 0.25 \\ 0.30 & 0.55 & 0.15 \end{bmatrix} \end{array}$$